

COSC 594

DM Composition (II)

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Outline

- RECAP

- DIFFUSION MODEL SOUP

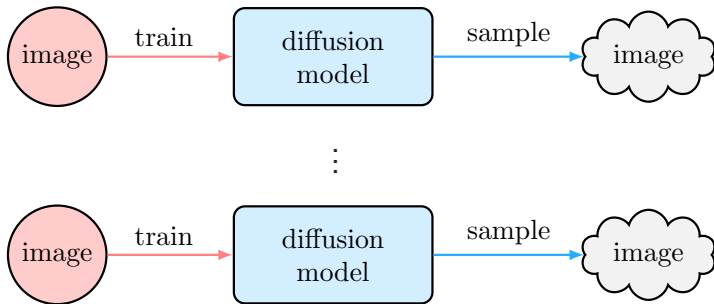
Outline

■ RECAP

Combination of DMs



⋮



$$x_0 \sim p^1(\cdot), \dots, p^K(\cdot)$$

How do we combine several pretrained DMs?

Diffusion model as energy-based model

Generative process of diffusion model: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}s_{\theta^*}(x_t, t) + \sigma(t)\epsilon_t$

Optimal score function: $s_{\theta^*}(x_t, t) = \nabla \log p(x_t)$

■ ENERGY-BASED DISTRIBUTION

$$p(x) \propto e^{-E(x)} \implies \nabla \log p(x) = -\nabla_x E(x)$$

★ Langevin dynamics

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla_x E(x_t) + \sigma(t)\epsilon_t$$

Score function as negative gradient of energy function

Diffusion model composition: conjunction

K diffusion models: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla \log p^k(x_t) + \sigma(t)\epsilon_t$ for $k = 1, \dots, K$

■ DISTRIBUTION CONJUNCTION

$$p^{\text{compose}}(x) \propto \prod_{k=1}^K p^k(x) \implies \nabla \log p^{\text{compose}}(x) = \sum_{k=1}^K \nabla \log p^k(x)$$

★ composed generative process

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}} \sum_{k=1}^K \nabla \log p^k(x_t) + \sigma(t)\epsilon_t$$

Conjunction via **adding up score functions**

Diffusion model composition: negation

K diffusion models: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla \log p^k(x_t) + \sigma(t)\epsilon_t$ for $k = 1, \dots, K$

■ DISTRIBUTION NEGATION

$$p^{\text{compose}}(x) \propto \frac{p^1(x)}{p^2(x)} \implies \nabla \log p^{\text{compose}}(x) = \nabla \log p^1(x) - \nabla \log p^2(x)$$

★ composed generative process

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}(\nabla \log p^1(x_t) - \nabla \log p^2(x_t)) + \sigma(t)\epsilon_t$$

Negation via **subtracting a score function**

Diffusion model composition: disjunction

K diffusion models: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}x_t + \frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla \log p^k(x_t) + \sigma(t)\epsilon_t$ for $k = 1, \dots, K$

■ DISTRIBUTION CONJUNCTION

$$p^{\text{compose}}(x) \propto \sum_{k=1}^K p^k(x) \implies \nabla \log p^{\text{compose}}(x) \neq \sum_{k=1}^K \nabla \log p^k(x)$$

★ normalization constant matters

$$\nabla \log p^{\text{compose}}(x) = \nabla \log \sum_{k=1}^K \frac{1}{Z^k} p^k(x)$$

Disjunction needs **mixture weights**

Outline

- MOTIVATION
- DIFFUSION MODEL SOUP

Linear approximation of fine-tuning

Score function: $s_{\theta}(x_t, t)$ w/ parameter θ

■ LINEAR APPROXIMATION

$$s_{\bar{\theta}+\delta}(x_t, t) \approx s_{\bar{\theta}}(x_t, t) + \nabla_{\theta} s_{\bar{\theta}}(x_t, t) \delta$$

$s_{\bar{\theta}}(x_t, t)$ – base diffusion model

δ – parameter perturbation

Fine-tuning updates a portion of network parameters

Diffusion model soup

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

Linear approximation: $s_{\theta^k}(x_t, t) = s_{\bar{\theta} + \delta^k}(x_t, t) \approx s_{\bar{\theta}}(x_t, t) + \nabla_{\theta} s_{\bar{\theta}}(x_t, t) \delta^k$

■ SOUP MODEL

$$\begin{aligned} s_{\theta^{\text{soup}}}(x_t, t) &:= s_{\sum_{k=1}^K w_k \theta^k}(x_t, t) \\ &= s_{\sum_{k=1}^K w_k (\bar{\theta} + \delta^k)}(x_t, t) \\ &\approx s_{\bar{\theta}}(x_t, t) + \nabla_{\theta} s_{\bar{\theta}}(x_t, t) \sum_{k=1}^K w_k \delta^k \\ &= \sum_{k=1}^K w_k \left(s_{\bar{\theta}}(x_t, t) + \nabla_{\theta} s_{\bar{\theta}}(x_t, t) \delta^k \right) \\ &\approx \sum_{k=1}^K w_k s_{\theta^k}(x_t, t) \end{aligned}$$

Diffusion model composition: conjunction

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(x_0)$ for $k = 1, \dots, K$

■ COMPOSED MODEL

$$p^{\text{compose}}(x_0) = \prod_{k=1}^K \left(p^k(x_0)\right)^{1/K} \times \frac{1}{Z}$$

$$\implies p(x_t) = \int p^{\text{compose}}(x_0) p(x_t | x_0) dx_0$$

$$\implies \nabla \log p(x_t) = \nabla \log \int p^{\text{compose}}(x_0) p(x_t | x_0) dx_0$$

Composing first, then scoring

Approximate score function

■ COMPOSED SCORE FUNCTION

$$\frac{1}{K} \sum_{k=1}^K \nabla \log p^k(x_t) = \nabla \log \prod_{k=1}^K \left(p^k(x_t) \right)^{1/K} = \nabla \log p'(x_t)$$

$$p \neq p'$$

★ composed generative process

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} x_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \frac{1}{K} \sum_{k=1}^K \nabla \log p^k(x_t) + \sigma(t) \epsilon_t$$

★ soup model

$$\frac{1}{K} \sum_{k=1}^K s_{\theta^k}(x_t, t) \approx s_{\theta^{\text{soup}}}(x_t, t) = s_{\frac{1}{K} \sum_{k=1}^K \theta^k}(x_t, t)$$

Composed score function $\equiv$ Score function of composed model

Diffusion model composition: disjunction

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(x_0)$ for $k = 1, \dots, K$

■ COMPOSED MODEL

$$p^{\text{compose}}(x_0) = \frac{1}{K} \sum_{k=1}^K p^k(x_0)$$

$$\implies p(x_t) = \int p^{\text{compose}}(x_0) p(x_t | x_0) dx_0$$

$$\implies \nabla \log p(x_t) = \nabla \log \int p^{\text{compose}}(x_0) p(x_t | x_0) dx_0$$

Composing first, then scoring

■ SCORE DECOMPOSITION

$$\begin{aligned}\nabla \log p(x_t) &= \nabla \log \int p^{\text{compose}}(x_0) p(x_t | x_0) dx_0 \\&= \nabla \log \left(\frac{1}{K} \sum_{k=1}^K p^k(x_t) \right) \\&= \frac{1}{\frac{1}{K} \sum_{k=1}^K p^k(x_t)} \frac{1}{K} \sum_{k=1}^K \nabla p^k(x_t) \\&= \frac{1}{\frac{1}{K} \sum_{k=1}^K p^k(x_t)} \frac{1}{K} \sum_{k=1}^K p^k(x_t) \nabla \log p^k(x_t) \\&= \sum_{k=1}^K \underbrace{\frac{\frac{p^k(x_t)}{K}}{\frac{1}{K} \sum_{i=1}^K p^i(x_t)}}_{\text{time-varying weight } w_k(x_t)} \nabla \log p^k(x_t)\end{aligned}$$

**Score function of composed model $\equiv$ Composed score function w/
time-varying weights**

Exact score function

■ COMPOSED SCORE FUNCTION

$$\sum_{k=1}^K w_k(x_t) \nabla \log p^k(x_t) = \nabla \log \left(\frac{1}{K} \sum_{k=1}^K p^k(x_t) \right)$$

★ composed generative process

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} x_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} \sum_{k=1}^K w_k(x_t) \nabla \log p^k(x_t) + \sigma(t) \epsilon_t$$

★ soup model

$$\sum_{k=1}^K w_k(x_t) s_{\theta^k}(x_t, t) \approx s_{\theta^{\text{soup}}}(x_t, t) = s_{\sum_{k=1}^K w_k(x_t) \theta^k}(x_t, t)$$

Composed score function $\equiv$ Score function of composed model

Creation of hybrid artistic styles

	A woman in white shirt holding up a cellphone.	A young boy is posing in a large grassy area.	A mountain goat stands on top of a rock on a hill.	People are holding wine glasses next to each other.	a man that has a shirt and a tie	a woman looking at a mirror while brushing her teeth	this woman is at a desk using a computer
SD2.1							
Ukiyo-e							
Romanticism							
Ukiyo-e & Romanticism							

Summary

- RECAP

- DIFFUSION MODEL SOUP

Conjunction (also negation), disjunction